

MLOPS ROADMAP



MACHINE LEARNING ENGINEER INTERVIEW CHEATSHEET (2025 Edition)

1. Foundations of ML (Refresh Quickly)

Topic	What You Must Know
Types of Learning	Supervised, Unsupervised, Semi-supervised, Reinforcement
Common Tasks	Classification, Regression, Clustering, Ranking



Data Preprocessing

Imputation, Scaling (`StandardScaler` , `MinMaxScaler`)
Encoding

Splitting Data

`train_test_split` , Stratified K-Fold, `TimeSeriesSplit`

Model Validation

Cross-validation, Grid/Random/Optuna Search

Bias-Variance Tradeoff

Underfit = High Bias; Overfit = High Variance

2. Must-Know ML Algorithms (With Use Cases)

Category	Algorithms	Use Case (2025-relevant)
Linear	Linear/Logistic Regression	Simple tabular datasets
Tree-based	Random Forest, XGBoost, LightGBM	Tabular, high-performance models
Distance	KNN, DBSCAN	Recommenders, anomaly detection
Dim. Red.	PCA, t-SNE, UMAP	Preprocessing + visualization
Ensemble	Bagging, Boosting, Stacking	Kaggle/tabular problems
Deep Learning	CNN, RNN, Transformers	NLP, Vision, Time Series

 2025 Trend: **Transformer-based tabular models** (TabNet, FTTransformer) are gaining adoption in industry.

3. Evaluation Metrics Cheat Table

Problem	Metrics to Know
Classification	Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC
Regression	RMSE, MAE, R^2 , MAPE
Imbalanced Data	F1, ROC-AUC, PR-AUC

 2025 Update: PR-AUC is often preferred over ROC-AUC in imbalanced data scenarios.

4. Tooling & Libraries (Must-Have)

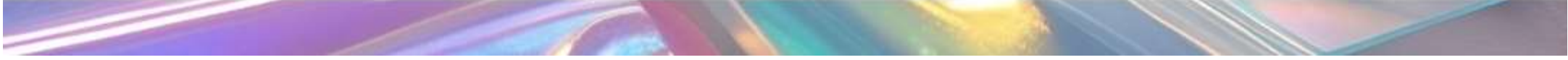
Task	Tools to Know (2025)
Model Building	Scikit-learn, XGBoost, LightGBM, CatBoost
Deep Learning	PyTorch  , TensorFlow/Keras
Experiment Tracking	MLflow, Weights & Biases
Model Deployment	FastAPI, Docker, TorchServe, BentoML
Feature Store	Feast, Tecton

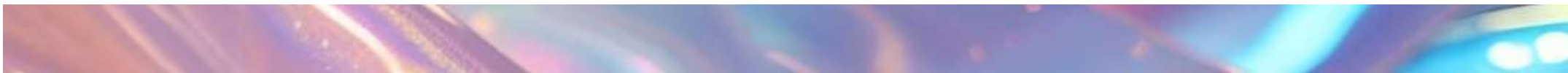


✿ 6. ML System Design – 2025 Hot Topics

✓ Be ready for open-ended ML design questions like:

🦷 Sample System Design Questions:

- How would you build a real-time fraud detection system?
 - Design a machine learning pipeline for churn prediction.
 - How would you handle continuous learning / model drift?
 - Build a recommendation engine for a music app.
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Must Discuss:

- Data ingestion (streaming or batch)
 - Offline vs Online models
 - Feature engineering & storage
 - Deployment: REST API, latency, scalability
 - Model retraining & monitoring
 - Experiment tracking
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7. Behavioral Interview (STAR Format)

Question

Tip

"Tell me about a time you failed."

Show learning & recovery.

"Describe an ML project end-to-end."

Structure: problem → data → model → results → impact.

"How do you work with cross-functional teams?"

Mention communication + business alignment.



2025 Bonus: Be prepared to talk about **AI ethics**, **model bias**, and **responsible AI**.